Do Now · retrieval

Lesson 092 · The Three Buses: City Highways

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What did PC advance to after sequential instruction at20?

Q2

What is MDR used for?

Q3

Which unit emits control signals?

Q4

Convert binary 10110000_2 to hexadecimal.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Address bus identifies the location the processor wants to access; typically the CPU drives address lines on a simple memory read.

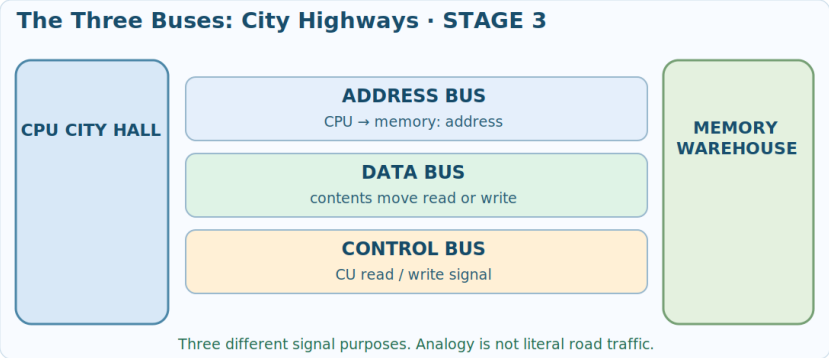
Core idea 2

Explain — Data bus transfers the contents between memory and processor; for ordinary memory reads the bits travel memory-to-CPU.

Core idea 3

Evaluate — Control bus signals operation and coordination, including read/write; distinguish purpose and direction rather than treating all buses as one road.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

address bus — Identifies location.

data bus — Transfers content bits.

control bus — Signals requests/status.

memory read — Retrieve contents of addressed location.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

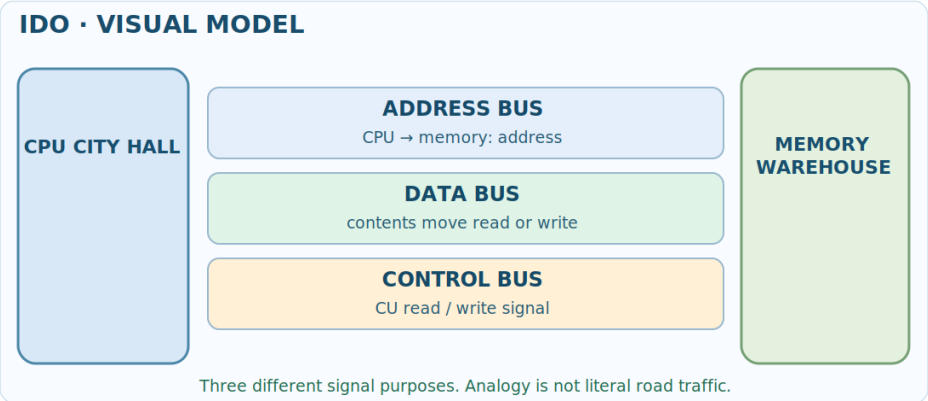
Problem

CPU reads memory[12]=99. Label each bus with what it carries.

Follow the worked method

1. Explain the mechanism — Address bus12; control bus READ; data bus returns99, placed in MDR.
— Address12 and content99 are different types of information.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the mechanism
Address bus44, data bus25 from CPU to memory, control bus WRITE.
On writing, data flows towards memory while address still identifies destination.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A read from address7 returns value150. Identify address/data/control signals. [6]

Question 2

Explain role and typical direction of all three buses for a write to memory address20 of value9. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate why confusing addresses with contents can break CPU fetch models; use PC,MAR,MDR,CIR in example. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The address bus and data bus always carry the same binary number.

Use key vocabulary from memory

1. Explain address bus in your own words.

2. Explain data bus in your own words.

3. Explain control bus in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.